Portfolio Risk–Return Analysis of Selected JSE Stocks

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Date: February 2026
Objective: Evaluate historical performance and identify the optimal risk-adjusted portfolio allocation.

Executive Summary

This analysis evaluates five JSE-listed equities between 2018 and 2025 to assess return, volatility, and diversification characteristics.

Key objectives:

- Compare individual stock performance
- Measure risk using standard deviation
- Analyse correlations between assets
- Simulate portfolio allocations
- Identify the portfolio with the highest Sharpe ratio

The goal is to determine whether diversification improves risk-adjusted returns relative to holding a single asset.

Summary of Findings

- SHP.JO delivered the strongest risk-adjusted performance
- SOL.JO exhibited highly disproportionate volatility
- Optimal portfolio heavily weighs on SHP.JO due to its superior risk-adjusted profile and allocates minimal capital to SOL.JO given its high risk and low return contribution.
- Max Sharpe portfolio: 19.3% return, 24% volatility

Business Problem

Investors must balance expected return against volatility.
Holding a single high-return stock may expose an investor to excessive risk.

This analysis explores whether combining assets across sectors (financials, retail, resources, and global tech exposure) improves portfolio stability and maximises risk-adjusted performance.

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import sankeyx as sex

plt.style.use('default')
```

In [2]:
```
#JSE Tickers
tickers = ["FSR.JO", "ASL.JO", "SHP.JO", "PRX.JO", "SOL.JO"]

#Analysis Period
start_date = "2018-01-01"
end_date = "2025-01-01"

#Downloading tickers
data = yf.download(tickers, start=start_date, end=end_date, auto_adjust = True)

prices = data["Close"] # "Close" in data.columns else data
```

```
[*********************100%***********************]  5 of 5 completed
```

Data Overview

The dataset contains daily adjusted closing prices for each selected stock. Adjusted prices account for dividends and stock splits, making them suitable for total return analysis.

In [3]:
```
prices.describe()
```

```
The history saving thread hit an unexpected error (OperationalError('attempt to write a readonly database')).History will not be written to the database.
```

Out[3]:

Ticker	ASL.JO	FSR.JO	PRX.JO	SHP.JO	SOL.JO
count	1526.000000	1526.000000	1526.000000	1526.000000	1526.000000
mean	5244.037444	6109.250875	60287.470399	29022.902363	2951.054329
std	1978.87998	1535.642104	19502.000797	5877.080735	8509.490432
min	2082.863281	3379.773440	31834.063659	16012.384664	2585.268646
25%	4280.530029	5477.398671	53063.999023	15868.205752	5400.010518
50%	5088.796875	6364.291321	60865.270287	27339.190664	2329.962625
75%	5950.018259	6737190518	67378.701772	34451.018024	2764.697441
max	8142.512438	8681.484375	88903.780002	39443.900708	42988.234375

Exploratory Data Analysis (EDA)

In [4]:
```
# Drop rows with missing price data to ensure consistent time alignment across all assets
prices.isna().sum()
prices = prices.dropna()

returns = prices.pct_change().dropna()

returns.describe()
```

Out[4]:

Ticker	ASL.JO	FSR.JO	PRX.JO	SHP.JO	SOL.JO
count	1525.000000	1525.000000	1525.000000	1525.000000	1525.000000
mean	0.000701	0.000381	0.000554	0.000446	0.000076
std	0.018016	0.012377	0.015501	0.016766	0.044391
min	-0.131594	-0.142791	-0.173925	-0.099922	-0.465657
25%	-0.010320	-0.010098	-0.010399	-0.009392	-0.019197
50%	0.000740	0.000668	-0.000039	-0.000035	-0.001077
75%	0.014769	0.010684	0.012680	0.009765	0.016343
max	0.198202	0.157758	0.279771	0.138302	0.554834

Average Annual Returns

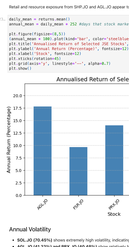
The average annual returns indicate differences in long-term performance across the selected JSE stocks.

- **SHP.JO (21.33%)** delivered the highest average annual return, suggesting strong historical growth over the selected period.
- **ASL.JO (17.7%)** demonstrated solid performance, likely reflecting its exposure to global commodity cycles.
- **PRX.JO (15.97%)** showed moderate growth that is consistent with its international tech exposure.
- **FSR.JO (9.61%)** produced steady but comparatively lower returns, typical of a large financial institution.
- **SOL.JO (-0.91%)** underperformed significantly compared to other assets. This indicates weak historical growth during the selected period.

Retail and resource exposure from SHP.JO and ASL.JO appear to have generated the strongest returns, while energy (SOL.JO) contributed the least.

In [5]:
```
daily_mean = returns.mean()
annual_mean = daily_mean * 252 #days that stock markets are open

plt.figure(figsize=(8,5))
annual_mean.plot(kind='bar', color='steelblue')
plt.title('Annualized Return of Selected JSE Stocks', fontsize=14)
plt.ylabel('Annual Return (Percentage)', fontsize=12)
plt.xlabel('Stock', fontsize=12)
plt.xticks(rotation=0)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



Annual Volatility

- **SOL.JO (70.45%)** shows extremely high volatility, indicating significant price fluctuations and high risk exposure.
- **ASL.JO (31.24%)** and **PRX.JO (40.6%)** show relatively high risk, reflecting sensitivity to global commodity and tech market movements.
- **FSR.JO (23.36%)** reflects moderate risk, consistent with the exposure in large-cap financial sectors.
- **SHP.JO (29.84%)** exhibits stable price behaviour, as shown in its lower volatility.

Financial and retail exposures provide lower risk profiles compared to energy and resource-linked assets.

In [6]:
```
daily_vol = returns.std()
annual_vol = daily_vol * (252 ** 0.5) # Volatility scales with the square root of time due to properties of variance under independent returns

plt.figure(figsize=(8,5))
annual_vol.plot(kind='bar', color='steelblue')
plt.title('Annualised Volatility of Selected JSE Stocks', fontsize=14)
plt.ylabel('Annual Volatility (Percentage)', fontsize=12)
plt.xlabel('Stock', fontsize=12)
plt.xticks(rotation=0)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```

Risk/Return Comparative Analysis

A comparison of annualised volatility and returns is critical to identify attractiveness of selected stocks.

- **SHP.JO (Return: 21.33%, Risk: 29.84%)** presents the most attractive standalone profile with the highest return and lowest volatility.
- **ASL.JO (Return: 17.76%, Risk: 41.32%)** maintains higher risk with solid return potential.
- **PRX.JO (Return: 13.81%, Risk: 40.68%)** demonstrates risk with moderate growth potential.
- **FSR.JO (Return: 9.61%, Risk: 33.36%)** moderate risk with steady, although lower growth.
- **SOL.JO (Return: 1.91%, Risk: 70.45%)** minimal return with severely high risk. This profile is the least attractive.

Key Observations

- SHP.JO is the most efficient asset over the selected period with an attractive risk/return ratio.
- SOL.JO increases portfolio risk significantly without a proportionate return contribution.
- Diversification benefits will depend on correlation structure rather than return alone.

Although individual asset performance is informative, an optimal portfolio construction requires analysing correlation and covariance to determine whether diversification can reduce risks and optimise returns.

Note: Volatility scales with the square root of the period due to properties of variance under independent returns.

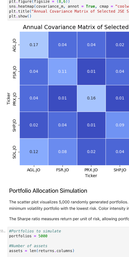
Asset Correlation

Diversification benefits arise when correlations between assets are less than 1.

In [7]:
```
asset_cor = returns.corr()

asset_cor.head()

plt.figure(figsize = (8,6))
sns.heatmap(asset_cor, annot = True, cmap = "coolwarm", fmt =".2f", linewidth = 0.5)
plt.title('Correlation Matrix of Stock Returns', fontsize = 14)
plt.show()
```



Covariance Matrix

In [8]:
```
covariance_m = returns.cov() * 252

plt.figure(figsize = (8,6))
sns.heatmap(covariance_m, annot = True, cmap = "coolwarm", fmt =".2f", linewidth = 0.5)
plt.title('Annual Covariance Matrix of Selected JSE Stocks', fontsize = 14)
plt.show()
```

Portfolio Allocation Simulation

The scatter plot visualizes 5,000 randomly generated portfolios. The red star represents the maximum Sharpe Ratio portfolio with the best risk-adjusted return, and the blue star represents the minimum volatility portfolio with the lowest risk. Color intensity indicates the Sharpe ratio which shows the risk-return trade-off and helps investors choose optimal allocations.

The Sharpe ratio measures return per unit of risk, allowing portfolios to be compared on a risk-adjusted basis rather than raw return alone.

In [9]:
```
#Portfolios to simulate
portfolios = 5000

#Number of assets
assets = len(returns.columns)

#Default risk-free rate
risk_rate = 0.0

results = np.zeros(3, portfolios)
weights_record = []

for i in range(portfolios):
    #Random weights
    weights = np.random.random(assets)
    weights /= np.sum(weights)

    #Return
    port_return = np.dot(weights, annual_mean)

    #Volatility
    port_vol = np.sqrt(np.dot(weights.T, np.dot(covariance_m, weights)))

    #Sharpe ratio
    sharpe = (port_return - risk_rate) / port_vol

    #Storing results
    results[0, i] = port_return
    results[1, i] = port_vol
    results[2, i] = sharpe
    weights_record.append(weights)

results_data = pd.DataFrame(results.T, columns = ["Return", "Volatility", "Sharpe"])
weights_df = pd.DataFrame(weights_record, columns = returns.columns)

portfolio_data = pd.concat([results_data, weights_data], axis = 1)

max_sharpe_index = results_data.loc[:, "Sharpe"].idxmax()
max_sharpe_folio = portfolio_data.loc[max_sharpe_index]
```

In [10]:
```
min_vol_index = results_data["Volatility"].idxmin()
min_vol_port = portfolio_data.loc[min_vol_index]

plt.figure(figsize=(12,7))

# All portfolios
plt.scatter(results_data['Volatility'], results_data['Return'], c=results_data['Sharpe'], cmap='viridis', alpha=0.6)
plt.colorbar(label='Sharpe Ratio')

# Maximum Sharpe ratio portfolio (red star)
plt.scatter(max_sharpe_folio['Volatility'], max_sharpe_folio['Return'], color='red', marker='*', s=300, label='Max Sharpe')

# Minimum volatility portfolio (blue star)
plt.scatter(min_vol_port['Volatility'], min_vol_port['Return'], color='blue', marker='*', s=300, label='Min Volatility')

# Labels and styling
plt.xlabel('Annualized Volatility (Risk)', fontsize=12)
plt.ylabel('Annualized Return', fontsize=12)
plt.title('Simulated Portfolio Optimization - Risk vs Return', fontsize=14)
plt.legend()
plt.grid(True, linestyle='--', alpha=0.5)
plt.show()

print("Maximum Sharpe Ratio Portfolio")
print(max_sharpe_folio.T)
print("Minimum Volatility Portfolio")
print(min_vol_port)
```

```
Maximum Sharpe Ratio Portfolio
Return    0.193454
Volatility  0.239811
Sharpe    0.806965
ASL.JO    0.292126
FSR.JO    0.000000
PRX.JO    0.218213
SHP.JO    0.489172
SOL.JO    0.000489
Name: 4315, dtype: float64
Minimum Volatility Portfolio
Return    0.132888
Volatility  0.163024
Sharpe    0.815365
FSR.JO    0.226248
PRX.JO    0.000000
SHP.JO    0.570365
SOL.JO    0.203387
Name: 1166, dtype: float64
```

The simulated portfolio with the maximum Sharpe ratio has an expected annual return of 19.55% with an annualised volatility of 23.60%. The optimal allocation heavily favours SHP.JO (48.81%) and ASL.JO (29.21%), while the remaining stocks have minimal allocations. This portfolio offers the best risk-adjusted return among the 5,000 simulated portfolios, balancing expected returns against volatility.

Conclusion

This analysis evaluated the historical performance of five JSE-listed equities between 2018 and 2025 using modern portfolio theory principles. By comparing annualised returns, volatility, correlations, and simulated portfolio allocations, several key insights emerged.

SHP.JO demonstrated the strongest individual risk-return profile, combining high returns with comparatively low risk. In contrast, SOL.JO exhibited high volatility with very low return contribution, making it less attractive on a standalone basis.

The Monte Carlo simulation of 5,000 random portfolios revealed that the maximum Sharpe ratio portfolio achieved an expected annual return of approximately 19% with volatility near 24%. The optimal portfolio allocated the majority of capital to SHP.JO, reflecting its superior risk-adjusted characteristics, while assigning minimal weight to higher-volatility, lower-efficiency assets.

This outcome reinforces a core principle of portfolio theory: optimal asset allocation depends not only on individual returns but on covariance structure and diversification benefits. Assets that improve overall portfolio efficiency receive higher weight; even if they are not the absolute highest-return assets.

Limitations

This analysis relies on historical data and assumes return distributions remain stable over time. Market regimes, macroeconomic conditions, and structural changes may alter future performance. Additionally, transaction costs, taxes, and liquidity constraints were not accounted for in the simulation.

Final Insight

The project demonstrates that disciplined quantitative analysis can transform raw market data into actionable portfolio insights. By focusing on risk-adjusted returns and diversification, investors can construct portfolios that are more efficient than any individual asset.